

European Put Options.

Usually, a right but NOT an obligation to SELL an underlying @ the strike price!

At time 0: Agreement:

- the underlying asset: $S(t), t \geq 0$
- the exercise date: T
- the strike / exercise price: K

The put premium $V_p(0)$ is paid by the put's buyer to the put's writer.

At time T:

- The put's owner has a right but not an obligation to sell for K .
- The put's writer is obligated to do what the put's owner decides.

The put owner's optimal behavior is:

IF $S(T) < K$, then exercise.

PAYOFF:
 $K - S(T)$

IF $S(T) \geq K$, then do not exercise.

0

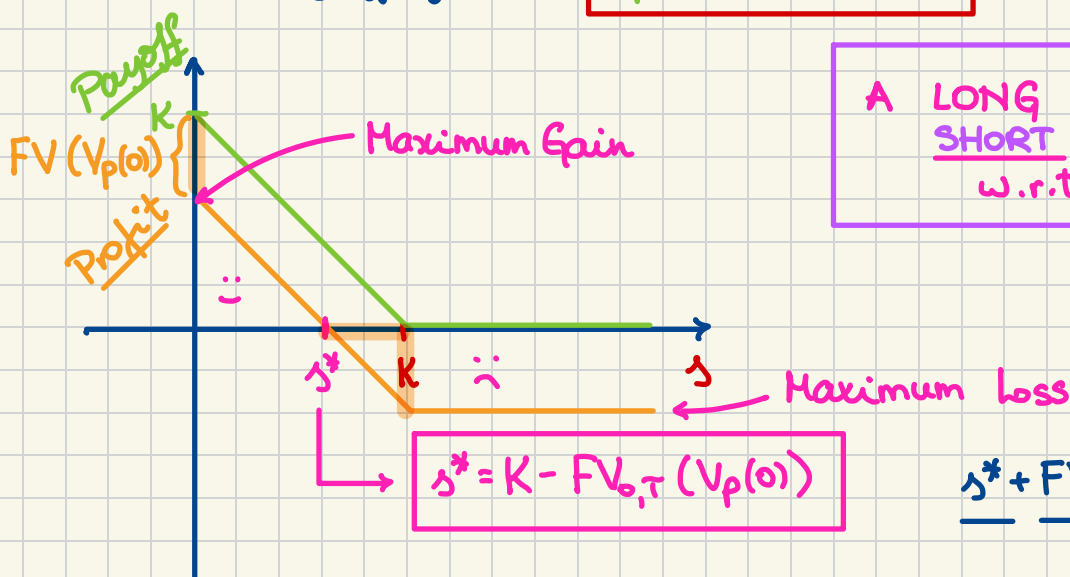
The payoff:

$$V_p(T) = \max(K - S(T), 0) = (K - S(T))_+$$

The payoff f'n:

$$v_p(s) = (K - s)_+$$

A LONG PUT IS A SHORT POSITION w.r.t. THE UNDERLYING.



Moneyiness.

Consider an option written @ time 0 w/ the exercise date T .



Imagine the **cashflow** that would happen if the option were exercised @ time t .

e.g.,

call: $S(t) - K$

put: $K - S(t)$

IF the **cashflow** is $\begin{cases} > 0, & \text{the option is in the money} \\ = 0, & \text{at the money} \\ < 0, & \text{out of the money} \end{cases}$

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IN-CLASS WORK 7

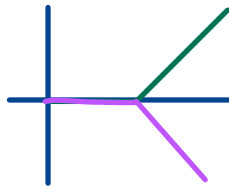
Introduction to Financial Mathematics

September 23, 2026

EUROPEAN PUT OPTIONS.

Problem 7.1. (2 points) In which of the following option positions is the investor exposed to an unlimited loss?

- ☒ a. Long put option.
- ☒ b. Short put option.
- ☒ c. Long call option.
- ☒ d. Short call option.
- ☐ e. None of the above.



Problem 7.2. The initial price of the market index is \$900. After 3 months the market index is priced at \$915. The nominal rate of interest convertible monthly is 4.8%. The premium on the put, with a strike price of \$930, is \$8.00. What is the profit at expiration for a long put?

- ☐ a. \$15.00 loss.
- ☐ b. \$6.90 loss.
- ☒ c. \$6.90 gain.
- ☒ d. \$15.00 gain.
- ☐ e. None of the above.

$$\text{Payoff: } (K - S(T))_+ = (930 - 915)_+ = 15$$
$$FV(V_p(0)) = 8(1.004)^3$$

$$15 - 8(1.004)^3 \approx 6.90 \quad \square$$

Problem 7.3. (3 points) The initial price of the market index is \$1000. After 3 months the market index is priced at \$950. The nominal rate of interest convertible quarterly is 4.0%. The premium on the long put, with a strike price of \$975, is \$10.00. What is the profit at expiration for this long put?

- a. \$12.00 loss
- b. \$14.90 loss
- c. \$12.00 gain
- d. \$14.90 gain
- e. None of the above.

Problem 7.4. Sample FM(DM) #12

Consider a European put option on a stock index without dividends, with 6 months to expiration, and a strike price of 1,000. Suppose that the nominal annual risk-free rate is 4% convertible semiannually, and that the put costs 74.20 now. What price must the index be in 6 months so that being long the put would produce the same profit as being short the put?

- ∴ i.r. a. \$922.83.
- b. \$924.32.
- ∴ PAYOFF c. \$1,000.00.
- ∴ CALL d. \$1,075.68.
- ∴ i.r. + CALL e. \$1,077.17.

The Break-even point:

$$S^* = K - FV_{0,T}(V_P(0))$$

$$= 1000 - 74.20(1.02) = \underline{924.32}$$



$$j = \frac{0.04}{2} = 0.02$$

Long Put: $(K - S(T))_+ - FV(V_P(0))$

Short Put: $-(K - S(T))_+ + FV(V_P(0))$

Problem 7.5. (3 points) *Source: Sample FM(DM) Problem #62.*

The stock price today equals \$100 and its price in one year is modeled by the following distribution:

$$S(1) \sim \begin{cases} 125 & \text{with probability } 1/2 \\ 60 & \text{with probability } 1/2 \end{cases}$$

The annual effective interest rate equals 3%.

Consider an at-the-money, one-year European put option on the above stock whose initial premium is equal to \$7.

What is the expected profit of this put option?

$$\mathbb{E}[g(x)] \neq g(\mathbb{E}[x])$$

$$\text{Payoff} : (100 - S(1))_+ \sim \begin{cases} 0 & \text{w/ prob. } 1/2 \\ 40 & \text{w/ prob. } 1/2 \end{cases}$$

$$\mathbb{E}[\text{Payoff}] = 20$$

$$\mathbb{E}[\text{Profit}] = 20 - 7(1.03) = 20 - 7.21 = 12.79$$

